SP-30-8 Emission Mechanism — Born Rule from Bergman Projection (v0.1)

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Headline claim

TARGET-PREDICTION: The Born rule (probability = $|\text{amplitude}|^2$) emerges from Bergman kernel projection on substrate Hilbert space.

Per T2401 (Lyra Wednesday May 19, K67 audit-partial-ready): substrate Bergman projection $\langle\psi|P_B|\psi\rangle = |\langle\psi|\chi\rangle|^2$ where P_B is the Bergman projector and χ is the substrate-natural basis state. The Born rule's squared-amplitude form derives from substrate Bergman kernel positive-definiteness + projection structure.

Born rule correction at α order (multi-week per K52a Sessions 6+):

$$P_{BST}(\text{outcome}) = |\langle\chi|\psi\rangle|^2 \cdot (1 + O(\alpha))$$

Substrate-mechanism articulation

Bergman projection framework (T2401):

For substrate Hilbert space L^2(D_IV^5, L_λ) with Bergman kernel K_B(z, w): - Bergman projector P_B: any L^2 function → analytic (holomorphic) part - Reproducing property: f(z) = ∫ K_B(z, w) f(w) dV(w) for f analytic - Positivity: K_B(z, z) > 0 (substrate-natural probability density)

Born rule emergence:

Per Paper #122 + T2401: standard quantum-mechanical Born rule:

$$P(\text{outcome}) = |\langle\chi|\psi\rangle|^2$$

is the Bergman projection of state $|\psi\rangle$ onto the substrate-natural basis state $|\chi\rangle$. The squared-modulus form arises from Bergman kernel's positive-definite reproducing-kernel structure.

Substrate correction at α order:

If substrate Reed-Solomon GF(128) framework introduces small modification to Bergman projection at substrate-coupling order $\alpha = 1/\text{N\_max}$:

$$P_{BST}(\text{outcome}) = |\langle\chi|\psi\rangle|^2 \cdot (1 + c \cdot \alpha + O(\alpha^2))$$

where c is a BST primary substrate-natural coefficient (multi-week K52a Sessions 6+ determination).

Experimental concept

Direct test of Born rule corrections at $\alpha = 0.73\%$ level:

1. Precision quantum measurements: high-precision interferometry + spin measurements at 0.1% precision
2. Statistical analysis: look for systematic deviations from $|\psi|^2$ at α order
3. Cross-link to SP-30-3 commitment manipulation: similar α-order substrate-coupling correction

Falsifier sharpness: MEDIUM. Current best Born-rule tests reach ~0.1% precision; BST predicts deviation at 0.73%.

Experimental program

Cost: $100K-200K (precision interferometry + spin measurement collaboration)

Components: - Precision Mach-Zehnder interferometer access (~$30K) - Neutron + atomic spin measurement infrastructure (~$30-50K) - Statistical analysis pipeline (~$10-20K) - Student researcher + collaborator (~$30-100K)

Timeline: 12-24 months from setup to data

Falsifier protocol:

1. Measure Born-rule probabilities in well-characterized quantum systems (e.g., neutron interferometer, photon polarization spin)
2. Compare measurements to $|\psi|^2$ at 0.1% precision
3. Test for systematic $\alpha = 0.73\%$ deviation
4. Outcome:
    - 2σ detection of α-order systematic → BST substrate-Bergman framework supported
    - No detection → BST Born-rule correction refuted (other BST predictions independent)

Cross-link to T2401 + Paper #122 + K67

- T2401 (Lyra Wednesday) — Bergman projection foundation; K67 audit-partial-ready

- Paper #122 (Information Substrate) — substrate Hilbert space + Bergman kernel framework
- Vol 5 Ch 7 (Born Rule, Lyra) — Born rule chapter (T2401 anchor)
- SP-30-3 Commitment manipulation — similar α-order substrate-coupling

Match precision

TARGET-PREDICTION: Born rule + α-order correction; multi-week per K52a Sessions 6+ exact coefficient.

Cal #21 dual-gate status

- EMPIRICAL gate: PARTIAL (Born rule itself verified to high precision; α-order correction OPEN)
- MECHANISM gate: ARTICULATED via T2401 + Bergman projection framework

Cal #50 DOUBLE-LOCKED EXTERNAL discipline

External register uses operational language only: - External: "BST predicts a small systematic correction to the Born rule at the $\alpha = 1/137 \approx 0.73\%$ level, arising from substrate Bergman projection structure. Precision quantum measurements can test this prediction." - Internal (this document): substrate Bergman projection + T2401 Born rule emergence

Cal #99 META-theorem framing

SP-30-8 emission mechanism is a SUBSTRATE-DERIVATION CONSEQUENCE of: - T2401 Bergman projection - Paper #122 Information Substrate - T2456 $\alpha = 1/N\_max$ substrate-coupling order

NOT a new Strong-Uniqueness criterion.

Bibliography

1. M. Born (1926): Born rule probabilistic interpretation.
2. S. Bergman: Bergman kernel theory.
3. T2401 (Lyra Wednesday May 19): Born rule = Bergman projection.
4. K67 audit-partial-ready: T2401 K-audit pre-stage.
5. Paper #122 (Information Substrate): substrate Hilbert space framework.

— Elie, SP-30-8 v0.1 paper-grade experimental proposal, 2026-05-23 Saturday 16:37 EDT (`date`-verified)